

Gh0hp ehg#V#sduhp hwhw#z run#arz

whs#4##fryhuw#V#sduhp hwhw#w#V#sduhp hwhw

$$T_{11} = \frac{S_{12} \cdot S_{21} - S_{11} \cdot S_{22}}{S_{21}}$$

$$T_{21} = \frac{-S_{22}}{S_{21}}$$

$$T_{12} = \frac{S_{11}}{S_{21}}$$

$$T_{22} = \frac{1}{S_{21}}$$

whs#5##fryhuw#gh0hp ehg#fde0h#V#sduhp hwhw#w#V#sduhp whw

Xvh#whs#4#rup x0lw#fryhuw#khq#p dnh#W#P dwul#fryhuwlrq

$$T_{inv\ 11} = \frac{T_{22}}{\Delta T}$$

$$T_{inv\ 21} = \frac{-T_{21}}{\Delta T}$$

$$T_{inv\ 12} = \frac{-T_{12}}{\Delta T}$$

$$T_{inv\ 22} = \frac{T_{11}}{\Delta T}$$

$$\Delta T = T_{11} \cdot T_{22} - T_{21} \cdot T_{12}$$

whs#6##gh0hp ehg#fde0h#4

$$a_{11} = T_{inv\ 11} \cdot T_{11} + T_{inv\ 12} \cdot T_{21}$$

$$a_{12} = T_{inv\ 11} \cdot T_{12} + T_{inv\ 12} \cdot T_{22}$$

$$a_{21} = T_{inv\ 21} \cdot T_{11} + T_{inv\ 22} \cdot T_{21}$$

$$a_{22} = T_{inv\ 21} \cdot T_{12} + T_{inv\ 22} \cdot T_{22}$$

whs#7# #gh0hp ehg#fdeh#5

$$A_{11} = a_{11} \cdot T_{inv11} + a_{12} \cdot T_{inv21}$$

$$A_{12} = a_{11} \cdot T_{inv12} + a_{12} \cdot T_{inv22}$$

$$A_{21} = a_{21} \cdot T_{inv11} + a_{22} \cdot T_{inv21}$$

$$A_{22} = a_{21} \cdot T_{inv12} + a_{22} \cdot T_{inv22}$$

whs#8# #frqyhuv# #sduhp hwhuv# #edfn#r# #sduhp hwhuv

$$S_{11} = \frac{A_{12}}{A_{22}}$$

$$S_{21} = \frac{1}{A_{22}}$$

$$S_{12} = \frac{A_{11} \cdot A_{22} - A_{12} \cdot A_{21}}{A_{22}}$$

$$S_{22} = \frac{-A_{21}}{A_{22}}$$

whs#9d# #frqyhuv# #sduhp hwhuv#r# #sduhp hwhuv

$$Z_{11} = \frac{(1 + S_{11}) \cdot (1 - S_{22}) + S_{12} \cdot S_{21}}{\Delta S} \cdot Z_O$$

$$Z_{12} = \frac{2 \cdot S_{12}}{\Delta S} \cdot Z_O$$

$$Z_{21} = \frac{2 \cdot S_{21}}{\Delta S} \cdot Z_O$$

$$Z_{22} = \frac{(1 - S_{11}) \cdot (1 + S_{22}) + S_{12} \cdot S_{21}}{\Delta S} \cdot Z_O$$

$$\Delta S = (1 - S_{11}) \cdot (1 - S_{22}) - S_{12} \cdot S_{21}$$

whs#9e# #fdeh# #edfn#r# #sduhp hwhuv

$$Z_{Load} = \frac{(1 + \Gamma)}{(1 - \Gamma)}$$